

1.

How many pointers need to be updated while inserting a node between two existing nodes in a doubly linear linked list?

- A. 2
- B. 4
- C. 1
- D. 0

Answer: B

2.

If a doubly linear linked list is traversed from the `head` to the last node, what must the `next` pointer of the last node be?

- A. NULL
- B. A pointer to the `prev` node
- C. A pointer to the `head`
- D. A pointer to the `tail`

Answer: A

3.

What is the worst-case time complexity for inserting a node at the beginning of a doubly linear linked list, given the head pointer?

- A. $O(n)$
- B. $O(1)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: B

4.

Which of the following is an advantage of using a doubly linked list over a singly linked list?

- A. Easier implementation for all operations.
- B. Less memory usage per node.
- C. Allows for efficient traversal in both forward and backward directions
- D. More efficient random access of elements.

Answer: C

5.

What is the worst-case time complexity for deleting a node from a specific position assuming head and tail are present in a doubly linear linked list?

- A. $O(n)$
- B. $O(1)$
- C. $O(n \log n)$
- D. $O(n^2)$

Answer: A

6.

If a doubly linked list contains nodes 112, 221, 454, 455, 554, 545, 121 and a pointer p points to node 554, then what would p->prev->next->next fetch?

- A. 121
- B. 545
- C. 554
- D. 454

Answer: B